

Resource-based carbon neutrality: Do resource-backed loans drive clean energy consumption in Africa south of the Sahara?

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Abstract

Achieving carbon neutrality in Sub-Saharan Africa (SSA) presents a paradox of abundance and constraints: vast natural resources coexist with limited access to sustainable finance. We empirically examine how resource-backed loans (RBLs) influence SSA's consumption of clean and renewable energy. Using panel data from 2000 to 2024 and applying Driscoll–Kraay and Hausman-Taylor robust models, the results reveal a complex picture of progress alongside ongoing structural challenges in SSA's path to carbon neutrality. We find that RBLs—though often justified as tools to finance infrastructure and development—have a dual nature. They increase access to clean cooking fuels and electricity but reduce renewable energy consumption. This indicates that RBLs, in their current form, may sustain carbon-intensive electrification rather than support a green transition. Therefore, shifting RBL frameworks toward renewable energy projects could create a strong synergy—expanding access while advancing decarbonization.

Keywords: resource-backed loans, clean energy consumption, access to electricity

1 Introduction

Carbon neutrality by 2050 represents a profound structural transformation of the global energy system, emphasizing investment in new infrastructure and efficiency improvements to shift from fossil fuels toward cleaner and more sustainable energy sources (Oteng et al., 2025; Oteng & Gamette, 2025). As Archer and Calvão (2025) note, this transition relies on efficient market mechanisms that drive the emergence of electrified, interconnected and low-carbon economies, powered by renewable technologies such as solar, wind and electric vehicles. However, transitioning to renewable energy comes at a significant financial cost. Global energy investment rose from US\$2.2 trillion in 2018 to US\$2.8 trillion in 2023, with the majority directed toward clean energy systems and technologies (Shi & Zhao, 2025), and clean energy investment alone has increased by approximately 40% since 2020 (Sharma et al., 2024). Under the net-zero emissions scenario, annual investment in clean energy is projected to more than triple from 2021 levels, reaching US\$4 trillion by 2030 (Shi & Zhao, 2025). Furthermore, the global market for renewable technologies, including wind turbines, solar panels, lithium-ion batteries and electrolyzers, is forecast to grow to US\$27 trillion by 2050, with renewable energy equipment such as battery technologies accounting for over 60% (Khattak, 2025). Yet, the regional disparities in this transition are striking: despite possessing nearly 60% of the world's highest-quality solar resources, Sub-Saharan Africa (SSA) accounts for only 1% of global installed solar photovoltaic capacity (Thapa et al., 2025). To achieve universal access to renewable energy by 2030, SSA needs approximately US\$25 billion annually, representing about 1% of global energy investment (Shi & Zhao, 2025).

However, SSA countries have inadequate resources from the public or private sectors (Coulibaly et al., 2022). Additionally, their access to capital markets, restricted over the last 20 years, has hindered many of them from the ability to better deploy resources on the financial

market (Mihalyi et al., 2022). Resource-rich nations use the potential product flows from their natural resource windfalls as loan collateral (Mihalyi & Scurfield, 2021; Mihalyi et al., 2022). Resource-backed loans (RBLs) are a financing type created during the 2000s natural resource booms (Kararach, 2025). RBLs are loans given to a government or public business that are repaid either directly with natural resources (in kind), like minerals or oil, or with the money made from those resources; alternatively, the loan may be secured by future natural resource revenues, or the reserves of natural resources may be used as collateral (Mihalyi et al., 2020).

From the total 52 RBLs signed globally between 2004 and 2018, 30 were signed by African nations, with the remaining 22 signed by Latin American nations (Coulibaly, 2024). RBLs increased from zero in 2006 to around 10 in 2018 and peaked at about 12 in 2021, as indicated in Figure 1. Following 2021, the number decreased somewhat before levelling off at about 10 RBLs by 2024. This displays a spike in lending activity in the late 2000s and early 2010s, which was followed by a decline in 2021. Between 2004 and 2018, the global RBLs valued US\$164 billion, with US\$66 billion going to SSA economies and US\$98 billion to Latin American nations (Coulibaly et al., 2022). Furthermore, the more comprehensive loan composition shows that Guinea received the most RBLs in Africa in 2017 (Mihalyi et al., 2020). The nation and a group of Chinese institutions inked a US\$20 billion RBL in 2017 (about 200% of the nation's GDP) (Mihalyi et al., 2020). In 2006, the Republic of Congo signed a US\$1.6 billion RBL, equal to 21% of its GDP, and made it the second-largest receiver of RBLs in Africa. The Sicomines infrastructure deal was signed with China in 2008. A total of US\$3 billion, or 16% of the nation's GDP, was borrowed (Brautigam, 2022).

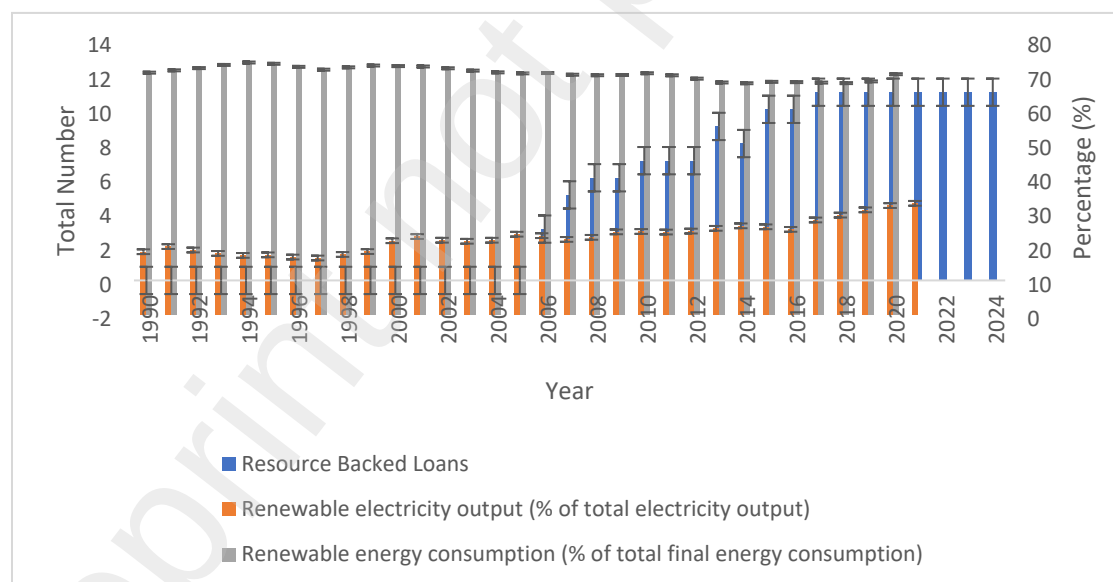


Figure 1: Trends of RBLs, electricity output and renewable energy consumption in SSA
Source: Authors' work

Understanding the implications of RBLs for Africa's energy transition remains an unresolved empirical question. Although RBLs have become increasingly prominent in financing

infrastructure and extractive projects, their theoretical and practical consequences for sustainable energy development are still poorly understood. Existing studies have examined RBLs from several perspectives, including deforestation (Coulibaly, 2025), credit ratings (Coulibaly et al., 2025), ecological efficiency (Coulibaly, 2024), sovereign default risks (Coulibaly, 2024), loan characteristics (Mihalyi et al., 2022), climate impacts (Coulibaly, 2023), and debt sustainability (Coulibaly et al., 2022). Other strands of research have focused on macroeconomic, institutional, or technological drivers of clean energy adoption (Murtaza & Luqman, 2025; Kararach, 2025). However, none of these studies has assessed how RBLs shape clean energy access or renewable energy consumption in SSA, a region characterized by persistent energy poverty, limited domestic capital, and strong reliance on externally financed infrastructure.

This study provides an empirical investigation of the impact of RBLs on clean energy access and renewable energy consumption in SSA. While earlier research has linked RBLs to environmental and macroeconomic outcomes, the energy-transition implications of these contracts remain entirely unexplored. By focusing on energy access—both electricity and clean cooking—and on renewable energy use, this study addresses a critical gap in the literature and extends the scope of inquiry beyond ecological efficiency and deforestation outcomes. We employ the Driscoll–Kraay estimator to correct for cross-sectional dependence, serial correlation, and heteroskedasticity that commonly affect multi-country panels subject to global shocks such as energy price fluctuations or shifts in international lending practices. Moreover, we operationalize clean energy using both access and consumption measures, enabling a more holistic assessment of the energy transition.

We examine the impacts of RBLs on energy-transition indicators in SSA countries from 2000 to 2024. The analysis focuses on three dimensions: (i) clean energy access (electricity and clean cooking), (ii) renewable energy consumption, and (iii) hydropower electricity generation. We seek to answer the following questions:

1. do RBLs influence clean energy access in SSA?
2. do RBLs affect renewable energy consumption in SSA?

The study makes a significant contribution to both the RBL and energy-transition literatures by conceptualizing RBLs as a form of externally financed man-made capital formation and demonstrating how they shape structural change within the energy sector. The multidimensional empirical approach—considering energy access, renewable energy use, and hydropower—offers a more comprehensive picture of transition pathways in resource-dependent economies. The findings contribute to ongoing policy debates on the developmental value of RBL-financed infrastructure and clarify the conditions under which these loans may either facilitate or hinder progress toward sustainable energy systems. In doing so, the study provides evidence relevant to long-term regional strategies aligned with global low-carbon and carbon-neutrality objectives.

The remaining part of the study proceeds as follows. The second section presents the methodology and the data sources. Section 3 presents the results and findings. Section 4 provides the conclusion, implications, and policy recommendations.

2 Data and Methodology

2.1 Estimation Strategy

The main objective of our study is to estimate the effect of resource-backed loans (RBLs) on energy transition. We employ the Driscoll-Kraay standard errors estimation approach. Considering the following:

$$y_{it} = \alpha_i + \beta' X_{it} + \varepsilon_{it}, i = 1, 2, \dots, N, t = 1, 2, \dots, T \quad 1$$

where y_{it} is the clean energy, including clean cooking access, renewable output, and access to electricity for country i at time t ; X_{it} is a $k \times 1$ vector of regressors including resource-backed loans, environmental sustainability, stability, and covariates; α_i captures unobserved country-specific fixed effects; ε_{it} is the idiosyncratic error term. After eliminating fixed effects (via within transformation), the standard fixed effects (FE) estimator is given by:

$$\hat{\beta}_{FE} = (X'MX)^{-1}X'My \quad 2$$

where $M = I_T - \frac{1}{T}\mathbf{1}_T\mathbf{1}_T'$ is the within-transformation matrix that removes individual means. However, under heteroskedasticity, serial correlation and cross-sectional dependence, the conventional variance estimator $\widehat{Var}(\hat{\beta}_{FE}) = \sigma^2(X'MX)^{-1}$ is no longer consistent. To address this, Driscoll and Kraay (1998) proposed a nonparametric covariance matrix estimator that is robust to general spatial and temporal dependence forms. The Driscoll-Kraay variance-covariance estimator is computed as:

$$\hat{V}_{DK} = (X'X)^{-1} \left(\sum_{l=-L}^L w_l \hat{\Gamma}_l \right) (X'X)^{-1} \quad 3$$

where L is the truncation lag parameter determining the number of autocorrelation terms to be considered; $w_l = 1 - \frac{|l|}{L+1}$ is the Bartlett kernel weight applied to each lag l ; $\hat{\Gamma}_l$ denotes the lag- l covariance estimator of the moment conditions:

$$\hat{\Gamma}_l = \frac{1}{T} \sum_{t=l+1}^T \left(\sum_{i=1}^N X'_{it} \hat{u}_{it} \right) \left(\sum_{i=1}^N X'_{i,t-l} \hat{u}_{i,t-l} \right)' \quad 4$$

with $\hat{u}_{it} = y_{it} - X'_{it}\hat{\beta}_{FE}$ representing the residuals from the fixed effects regression.

Driscoll-Kraay variance-covariance estimator allows unequal variances across cross-sectional units and accounts for serial correlation within units over time. It is even consistent when residuals are correlated across countries and asymptotic validity, thus, consistent as $T \rightarrow \infty$ for fixed N . Thus, the resulting Driscoll-Kraay estimator of $\hat{\beta}$ and its standard errors is expressed:

$$\hat{\beta}_{DK} = \hat{\beta}_{FE}, \text{Var}(\hat{\beta}_{DK}) = \hat{V}_{DK} \quad 5$$

This adjustment ensures that statistical inference remains valid under spatial and temporal dependence—conditions particularly relevant for SSA, where countries often share regional policy shocks, energy market linkages, and climatic conditions.

We conducted the cross-sectional dependence test to test whether a shock in one country affects others based on the Pesaran (2015; 2021) (CD) and cross-sectional dependence with power enhancement from Fan et al. (2015) (CDw+). Appendix A shows statistically significant CD for most variables, including renewable energy consumption, access to clean fuels and electricity

(both urban and rural), resource rents, and economic indicators ($p < 0.01$). Only political stability and the absence of violence indicate relatively country-specific governance and institutional stability behavior. The strong evidence of interdependence across panels underscores the need to employ Driscoll–Kraay's robust standard errors, which can adequately account for cross-sectional dependence and yield consistent parameter estimates.

2.2 Robustness checks

We employ the Hausman–Taylor (HT) estimator for robustness checks to analyze the effect of resource-backed loans on energy transition. The model is specified as:

$$Y_{it} = \alpha + X_{it}\beta + Z_i\gamma + u_i + \varepsilon_{it} \quad 6$$

where Y_{it} denotes energy transition (e.g., access to clean cooking, electricity production from renewable and renewable energy consumption); X_{it} represents time-varying variables such as RBL, resource rent, and macroeconomic indicators; Z_i includes time-invariant factors such as country; u_i is the unobserved country-specific effect; and ε_{it} is the idiosyncratic error term. Following Hausman and Taylor (1981), regressors are partitioned into exogenous and endogenous components:

$$X_{it} = [X_{1it}, X_{2it}], Z_i = [Z_{1i}, Z_{2i}] \quad 7$$

where some variables may be correlated with u_i . The HT estimator corrects for this endogeneity by using internal instruments—the within variation in X_{1it} to instrument X_{2it} , and the between variation in X_{1it} and Z_{1i} to instrument Z_{2i} .

This approach allows consistent estimation while retaining time-invariant variables. The expected sign of the RBL coefficient (β_p) is negative for energy transition. This is because resources such as crude oil and minerals are mostly used to secure loans and might limit the transition to renewables. The data, presented in Table 1, exhibit substantial cross-country (between) variation compared to within-country (temporal) variation, which supports using HT as a robustness check.

2.3 Data description

Our focus variable is a dummy variable indicating RBL adoption. Our panel data covers the period from 2000 to 2024. Data on RBLs is obtained from Mihályi et al. (2020), providing the most reliable and comprehensive evidence on such loans in developing countries. The dataset, compiled by Mihályi et al. (2020), integrates the China–Africa Loans Database (CARI, Johns Hopkins SAIS) and the China–Latin America Finance Database (Inter-American Dialogue and Boston University) and is further complemented by literature-based updates. Although the opacity of these transactions implies that some RBLs may not be fully disclosed (Rivetti, 2022), their study remains the most extensive source for Africa and Latin America. Based on this information, we constructed a binary variable equal to 1 if country i in year t contracted an RBL, and 0 otherwise (Coulibaly, 2025).

The dependent variables are renewable energy consumption, the percentage of the population with access to electricity and clean cooking fuels, in line with recent studies (Abba Yadou et al., 2024; Acheampong et al., 2025; Martey et al., 2022). Clean cooking is inaccessible to one-third of the world's population. Without access to clean cooking, 81900 premature deaths occur every year (IEA, 2025). One of the main causes of illnesses and deaths among women and children in low- and middle-income countries is still the lack of clean cooking energy and technologies. About 1 billion people in SSA lack access to clean cooking, which affects four of every five homes (IEA, 2025). Our main regressor is RBL. SDG 7.1 indicators that increasing financial assistance can promote improved access to modern energy. The data was gathered from diverse sources, including the World Bank's World Development Indicators (WDI) and Quality of Government Institute (QGI) ([QoG Data](#)) published by Jan et al. (2025). Appendix A provides the sources, measurements, and definitions of the variables. The selected SSA countries are provided in Appendix B.

3 Results and discussion

3.1 Descriptive statistics

The descriptive statistics are presented in Table 1 over time, within and between countries. Renewable energy consumption averages 67.1%, showing substantial variation across countries (between SD=24.0) and over time (within SD=5.3), suggesting moderate temporal fluctuations but large cross-country differences. Access to clean cooking fuels has a low mean of 17.3%, with high between-country variation (SD=23.4) and limited within-country change (SD=5.0), indicating persistent disparities across countries. Access to electricity averages 37.9% showing large between-country differences (SD=22.7) and some within-country improvement over time (SD=10.5).

Table 1: Descriptive Statistics

Variable		Mean	Std. dev.	Min	Max	Obs
Renewable energy consumption	Overall	67.083	24.171	5.6	97	N=979
	Between		24.010	11.939	96.364	n=44
	Within		5.342	42.958	101.244	T=22.25
Access to clean cooking fuels	Overall	17.263	23.723	0	98.1	N=1012
	Between		23.439	0.074	96.943	n=44
	within		5.034	-11.544	47.056	T=23
Access to electricity	overall	37.909	24.747	2.8	99.5	N=1042
	between		22.716	4.812	99.308	n=44
	within		10.520	10.010	79.480	T=23.682
RBL	overall	0.155	0.363	0	1	N=1100
	between		0.286	0	0.84	n=44
	within		0.227	-0.685	0.835	T=25
Resource rent	overall	11.712	10.865	0.009	51.844	N=951
	between		9.842	0.030	39.423	n=44
	within		5.353	-13.227	39.429	T=21.614

EnergyGDP	overall	11.376	7.496	2.228	44.891	N=671
	between		10.799	3.356	44.891	n=34
	within		2.693	0.281	32.241	T=19.735
Sustainability	overall	3.167	0.591	1	4.5	N=756
	between		0.496	1.346	3.85	n=39
	within		0.368	1.917	4.342	T=
						bar=19.385
Stability	overall	-0.571	0.885	-2.474	1.064	N=1002
	between		0.815	-2.139	1.019	n=44
	within		0.391	-2.106	0.880	T=22.773
Education	overall	41.862	22.091	6.867	97.075	N=574
	between		17.919	13.528	89.233	N=43
	within		10.612	12.860	99.133	T=
						bar=13.349

RBL has a mean of 0.155, reflecting that just 15.5% secured loans, but variation exists both across and within countries. Total natural resource rent averages 11.7% of GDP, with substantial between-country variation (SD=9.84) and moderate within-country movement, implying differences in resource dependence. GDP per unit of energy (EnergyGDP) has a mean of 11.4%, but strong between variation (SD=10.8) relative to within variation (SD=2.7), indicating structural differences among economies. Environmental sustainability rating averages 3.17, with moderate variation between and within countries, reflecting gradual policy or institutional differences. Political stability averages -0.57, with most of the variation occurring between countries, showing relatively stable but differing country conditions over time. The mean of education is 41.9, with considerable between-country variation (SD=17.9) and moderate within-country change (SD=10.6), reflecting both structural and temporal differences in education outcomes.

3.2 Impact of RBL on access to clean cooking

Table 2 presents the impact of RBL on access to clean cooking fuels and technologies in SSA. The empirical results from both the fixed effects (FE) and random effects (RE) estimations, as well as their Driscoll–Kraay robust counterparts, reveal consistent and theoretically coherent relationships between key macroeconomic and institutional variables and access to clean cooking fuels across SSA. The coefficient of RBL is positive and statistically significant at the 1% level across all model specifications, suggesting that countries utilizing such financing mechanisms experience higher levels of access to clean cooking fuels. Quantitatively, the FE estimates indicate that RBLs are associated with an approximate 3.8 to 4.3 percentage-point increase in access. This finding implies that resource-backed financing may facilitate infrastructure investments or social programs that enhance household access to modern cooking technologies. Such a relationship aligns with recent evidence that targeted external finance can bridge energy access gaps when effectively managed and directed toward inclusive sectors (see IEA, 2025; IMF, 2023; World Bank, 2022). Our results corroborate that of Twumasi et al. (2020) that credit accessibility significantly positively influences clean energy for cooking.

Education also demonstrates a positive and highly significant relationship across all models, reinforcing the notion that human capital development enhances clean energy adoption. The estimated coefficient (≈ 0.28) implies that improvements in education outcomes substantially raise the likelihood of access to clean cooking fuels. Higher levels of education increase clean cooking energy since they might be aware of the consequences and health implications of non-clean energy use. This finding corroborates prior studies emphasizing the role of education in shaping energy preferences, awareness of health impacts, and the affordability of clean technologies (Ouedraogo, 2017; Pachauri et al., 2021; Twumasi et al., 2020). By contrast, total natural resource rent has a significant negative impact on access to clean cooking. The coefficient (-0.23 to -0.31) indicates that higher resource dependence is associated with reduced clean cooking penetration, consistent with the resource curse hypothesis, where economies reliant on extractive industries exhibit weak diversification and limited investment in social infrastructure (Auty, 2001; Ploeg, 2011).

Table 2: RBLs and clean cooking energy access

	FE	RE	FE_Driscoll-Kraay	RE_Driscoll-Kraay
RBL	3.814*** (1.369)	4.298*** (1.358)	3.814* (2.041)	4.298* (2.076)
EnergyGDP	-0.381 (0.299)	0.095 (0.226)	-0.381* (0.196)	0.095 (0.154)
Education	0.276*** (0.042)	0.280*** (0.042)	0.276*** (0.069)	0.280*** (0.075)
Resource rent	-0.312*** (0.087)	-0.233*** (0.084)	-0.312*** (0.052)	-0.233** (0.088)
FDI	-8.63e-10*** (3.03E-10)	-9.96e-10*** (3.03E-10)	-8.63e-10* (4.11E-10)	-9.96e-10** (3.68E-10)
Political stability	0.023 (1.012)	0.087 (0.994)	0.023 (0.768)	0.087 (0.677)
Environmental sustainability	1.296 (0.980)	0.824 (0.969)	1.296 (0.826)	0.824 (0.748)
Block(base=Central Africa)				
East Africa		-7.174 (8.708)		-7.174 (14.56)
Southern Africa		-0.289 (10.82)		-0.289 (8.531)
West Africa		-0.861 (7.933)		-0.861 (8.228)
_cons	5.52 (3.949)	4.613 (7.503)	5.520** (2.321)	4.613 (5.572)
N	234	234	234	234

Note: Standard errors are in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Surprisingly, in Table 2, FDI also bears a negative and statistically significant coefficient, suggesting that much of the FDI inflow is concentrated in capital-intensive sectors such as oil, gas, or mining, with limited spillover effects on household energy access. This result underscores the importance of aligning FDI with inclusive development and clean energy objectives (UNCTAD, 2022). Other explanatory variables—including GDP per unit of energy use, political stability and avoidance of violence, and environmental sustainability—are statistically insignificant in most specifications, implying that their effects on clean cooking access may operate indirectly through structural transformation, governance quality, or policy coordination. Similarly, the regional dummy variables are not significant, indicating that variations in clean cooking access are more strongly driven by country-specific factors than by broad regional characteristics.

The consistency of signs and magnitudes across both conventional and Driscoll–Kraay robust estimators confirms the robustness of the results to heteroskedasticity and cross-sectional dependence. These findings collectively suggest that access to resource-backed financing and investments in education are critical pathways for accelerating progress toward SDG 7 of providing universal access to affordable, reliable, and modern energy, whereas overreliance on natural resource rents and unaligned FDI flows may hinder the achievement of SDG 7.

3.3 Impact of RBL on electricity accessibility

We present the impact of RBL on access to electricity. The empirical results presented in Table 4 evaluate RBL on electricity accessibility using FE, RE, and Driscoll–Kraay robust models to control for cross-sectional dependence and serial correlation. The coefficients of RBL are positive across all specifications, ranging from 1.495 to 1.693, but are statistically insignificant. This suggests that, although the effect is not statistically different from zero, RBLs have the potential to enhance access to electricity through infrastructure and energy sector investments. The statistical insignificance might reflect inefficiencies in the allocation of RBL toward non-electricity access-related projects. Our results align with previous research highlighting the governance and transparency challenges associated with resource-backed financing, which often limit its developmental impact (Mihalyi et al., 2022; Brautigam & Hwang, 2016).

GDP per energy use shows a positive coefficient in all models (0.240–0.883) and achieves statistical significance at the 10% level in the FE Driscoll–Kraay model. This finding implies that as the energy sector's share in economic output increases, access to electricity tends to improve. It underscores the importance of expanding and modernizing the energy sector as a pathway to broader electrification. Similar evidence from IEA (2022) suggests that improvements in energy productivity and sectoral efficiency contribute directly to electricity availability and affordability. Education demonstrates a consistently positive and highly significant effect ($p < 0.01$) across all model specifications. This strong relationship indicates that higher educational attainment significantly enhances access to electricity, possibly by increasing awareness of modern energy benefits, improving household income levels, and promoting the adoption of energy-efficient technologies. This result is consistent with findings by Lee et al. (2016) and Sadorsky (2013), who

argue that human capital development facilitates infrastructure utilization and accelerates energy transition processes.

Table 3: RBLs and electricity access

	FE	RE	FE_Driscoll-Kraay	RE_Driscoll-Kraay
RBL	1.495 (1.948)	1.693 (2.258)	1.495 (1.858)	1.693 (1.404)
EnergyGDP	0.883 (1.049)	0.240 (0.408)	0.883* (0.464)	0.240 (0.410)
Education	0.656*** (0.119)	0.682*** (0.102)	0.656*** (0.118)	0.682*** (0.114)
Resource rent	-0.285 (0.222)	-0.326* (0.184)	-0.285** (0.102)	-0.326** (0.143)
FDI	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	3.81e-10* (0.000)
Political stability	3.112 (2.199)	3.468** (1.702)	3.112* (1.580)	3.468* (1.870)
Environmental sustainability	1.874 (2.355)	2.120 (2.327)	1.874** (0.727)	2.120** (0.970)
Block(base=Central Africa)				
East Africa		-8.843 (6.466)		-8.843 (21.410)
Southern Africa		-23.32** (9.105)		-23.32* (12.770)
West Africa		-2.802 (7.114)		-2.802 (8.025)
_cons	-1.128 (13.010)	9.133 (9.900)	-1.128 (6.588)	9.133 (9.886)
N	234	234	234	234

Note: Standard errors are in brackets.*p<0.10,**p<0.05,***p<0.01.

Total natural resource rent exhibits a negative and statistically significant relationship with access to electricity in both the RE and Driscoll–Kraay models. This result supports the “resource curse” hypothesis, which posits that heavy dependence on natural resource rents can undermine structural transformation and inclusive infrastructure investment (Venables, 2016). The negative coefficient implies that resource-rich economies may underinvest in electricity infrastructure due to fiscal volatility, rent-seeking behavior, and weak institutional quality. However, FDI has a positive effect on access to electricity, yet its magnitude is nearly zero and only weakly significant in the RE Driscoll–Kraay model. This suggests that while FDI inflows can support energy infrastructure through technology transfer and capital accumulation, their overall impact on

electricity access remains minimal, likely because most foreign investments are directed toward extractive industries rather than power generation or distribution (Giroud, 2024).

Political stability exerts a positive and statistically significant effect on access to electricity in the RE and Driscoll–Kraay specifications. This highlights the critical role of stable political environments in promoting investment in energy infrastructure, ensuring consistent policy implementation, and attracting both domestic and foreign investors. Political instability, on the other hand, often disrupts electricity projects and discourages long-term commitments (Dirks & Schmidt, 2024). Environmental sustainability has a positive and significant impact under the Driscoll–Kraay estimations, suggesting that countries with stronger environmental performance and policy frameworks tend to achieve higher electricity access. This result may reflect the growing integration of renewable energy initiatives and environmentally conscious planning into national electrification strategies.

3.4 Effect of RBL on renewable energy consumption

We present the results of the effect of RBL on driving renewable energy consumption in Table 5. We estimated these results employing fixed effects, random effects, and Driscoll–Kraay robust estimators to address potential cross-sectional dependence and heteroskedasticity. The coefficient on BL is consistently negative and highly significant ($p < 0.01$) across all model specifications, ranging from -4.766 to -4.738 . This robust negative relationship indicates that an increase in RBL inflows is associated with a decline in renewable energy consumption. The result suggests that RBLs, often collateralized by future commodity revenues, may reinforce fossil fuel dependence rather than support renewable energy expansion. This finding is consistent with concerns that RBLs in Africa are disproportionately directed toward extractive and infrastructure sectors tied to oil, gas, and mining rather than clean energy investment (Mihalyi et al., 2022). Such financing structures might crowd out renewable energy development by locking countries into resource-dependent repayment obligations, thereby delaying green transition initiatives. Loans can be used and are often channeled into industries that emit higher carbon footprints, possibly increasing emissions through production and energy consumption. Our findings support Destek et al. (2025), who indicated that business loans drive carbon-intensive activities.

Table 4: RBLs and renewable energy consumption

	FE	RE	FE_Driscoll-Kraay	RE_Driscoll-Kraay
RBL	-4.766*** (0.854)	-4.738*** (1.58)	-4.766*** (1.216)	-4.738*** (1.013)
EnergyGDP	-0.678*** (0.187)	-0.709** (0.314)	-0.678** (0.275)	-0.709*** (0.184)
Education	-0.261*** (0.026)	-0.269*** (0.050)	-0.261*** (0.041)	-0.269*** (0.037)
Resource rent	0.078 (0.054)	0.073 (0.079)	0.078** (0.035)	0.073** (0.033)

FDI	-3.20E-11 (1.89E-10)	-8.74E-12 (2.63E-10)	-3.20E-11 (1.59E-10)	-8.74E-12 (1.75E-10)
Political stability	1.378** (0.631)	1.056 (1.153)	1.378** (0.599)	1.056* (0.549)
Environmental sustainability	-0.707 (0.611)	-0.645 (1.18)	-0.707 (0.451)	-0.645 (0.784)
Block(base=Central Africa)				
East Africa		6.931 (7.109)		6.931 (11.37)
Southern Africa		7.74 (7.043)		7.74 (6.392)
West Africa		-4.508 (8.627)		-4.508 (6.326)
_cons	94.45*** (2.463)	93.30*** (7.017)	94.45*** (3.022)	93.30*** (3.715)
N	234	234	234	234

Note: Standard errors are in brackets.*p<0.10,**p<0.05,***p<0.01.

GDP per energy use has negative and significant effects. This implies that many SSA economies remain dominated by fossil fuel–intensive energy sectors, where economic expansion in energy output reflects increased extraction and consumption of nonrenewable resources. The result aligns with studies showing that fossil-based industrialization continues to constrain the transition toward renewables in developing economies (IEA, 2022). It further highlights the structural inertia in SSA energy systems, where economic growth driven by hydrocarbons may impede the uptake of clean alternatives. Education has a significant negative impact at the 1% level in all models. This unexpected finding suggests that higher levels of education are associated with lower renewable energy consumption. One possible interpretation is that better-educated populations may have higher incomes and thus greater demand for modern but nonrenewable energy sources. While education contributes to overall energy access, it may initially reduce the relative share of renewables in the energy mix during the early stages of economic transition.

Resource rent exhibits a positive and statistically significant relationship with renewable energy consumption in the Driscoll–Kraay models. Thus, appropriate reinvestment of resource rent could finance renewable energy infrastructure and diversification programs. FDI has an insignificant effect across all models. This result may reflect the composition of FDI in SSA, which remains heavily skewed toward extractive sectors rather than renewable energy industries (Giroud, 2024). Without targeted incentives or green finance mechanisms, FDI is unlikely to drive substantial renewable energy adoption. Meanwhile, political stability has a positive and statistically significant relationship with renewable energy consumption in the FE and Driscoll–Kraay models. This indicates that stable political environments facilitate policy continuity, regulatory reforms, and investment confidence necessary for renewable energy expansion. Our findings support Dirks and Schmidt (2024). Stable governance structures enable governments to implement long-term renewable energy policies, attract clean technology investments, and

establish public–private partnerships. The coefficient on environmental sustainability is negative but statistically insignificant across all specifications. This suggests that, despite policy commitments, environmental sustainability indicators have yet to translate into measurable increases in renewable energy consumption. This gap could reflect implementation challenges, weak enforcement of environmental policies, or insufficient integration of renewable energy objectives into environmental frameworks.

3.5 Robustness of our findings

Our robustness results are presented in Table 6. The results indicate that RBLs significantly increase access to clean cooking fuels ($\beta=4.060, p<0.01$) but have an insignificant effect on electricity access and a strongly negative and significant impact on renewable energy consumption ($\beta=-6.285, p<0.01$). This finding suggests that although RBLs can finance infrastructure that improves household energy access—particularly, access to liquefied petroleum gas (LPG) and other modern cooking fuels—they are less effective in promoting renewable energy systems. Instead, such loans often channel resources toward fossil-fuel-based infrastructure or extractive sector projects, crowding out investments in renewables. This pattern underscores the “carbon financing paradox.” While RBLs help close short-term energy access gaps, they may inadvertently entrench fossil energy dependence, hindering long-term sustainability.

Table 6: Robustness of the effect of RBLs on energy transition

	Clean cooking fuels	Access to electricity	Renewable energy consumption
RBL	4.060*** (1.335)	1.646 (1.843)	-6.285*** (0.795)
EnergyGDP	-0.006 (0.244)	0.488* (0.286)	-0.851*** (0.116)
Education	0.277*** (0.041)	0.669*** (0.056)	-0.240*** (0.019)
Resources rents	-0.266*** (0.083)	-0.299*** (0.112)	-0.010 (0.053)
Political stability	0.014 (0.980)	3.127** (1.339)	2.016*** (0.512)
Environmental sustainability	0.898 (0.949)	2.076 (1.307)	
FDI	-0.000*** (0.000)	0.000 (0.000)	-0.000* (0.000)
Block	0.940 (3.491)	1.057 (3.187)	-6.541 (4.626)
_cons	1.396 (10.678)	-2.839 (10.323)	109.841*** (12.794)
sigma_u	20.102	18.031	25.121
sigma_e	4.079	5.673	2.735
Rho	0.960	0.910	0.988

<i>N</i>	234	234	298
Standard errors in parentheses. * $p<0.10$, ** $p<0.05$, *** $p<0.01$			

GDP per unit of energy use is positive and significant for electricity access ($\beta=0.488, p<0.1$) but negative for renewable energy consumption ($\beta=-0.851, p<0.01$). Education emerges as one of the most consistent and influential determinants across all energy outcomes. It exerts positive and statistically significant effects on clean cooking fuels ($\beta=0.277$) and electricity access ($\beta=0.669$), but a negative and significant effect on renewable energy consumption ($\beta=-0.240$). This duality aligns with the energy ladder hypothesis, which posits that as education and income levels rise, households tend to move away from traditional biomass toward modern—but often nonrenewable—energy sources. While this transition improves household welfare and health outcomes, it may temporarily reduce the share of renewables in the energy mix. Resource rents exhibit a negative and significant relationship with both clean cooking ($\beta=-0.266$) and electricity access ($\beta=-0.299$), while their effect on renewable energy consumption is statistically insignificant. However, political stability shows a strong and positive influence on both electricity access ($\beta=3.127, p<0.05$) and renewable energy consumption ($\beta=2.016, p<0.01$). yet, its effect on clean cooking fuels is statistically insignificant, implying that decentralized and community-level interventions in the cooking energy sector are less dependent on macro-political stability. For environmental sustainability, there is a positive but insignificant coefficient across models, indicating that while sustainability objectives are increasingly incorporated into national policy frameworks, they have yet to translate into tangible impacts on energy access or renewable energy use. Meanwhile, FDI presents mixed results—negative for clean cooking and renewable energy, and insignificant for electricity access—suggesting that FDI inflows remain concentrated in extractive industries rather than clean energy technologies.

3.6 Sensitivity analyses

Appendix C provides the results of the sensitivity analyses. In Southern and East Africa, RBLs appear to strongly enhance access to clean cooking fuels, but in Central and West Africa, RBLs do not significantly influence household energy access. For access to electricity, in Southern Africa, RBLs may be effectively financing energy generation and transmission infrastructure, supporting electrification efforts. West Africa's moderate effect indicates some success in using RBLs for power access expansion, possibly through hydro or gas-to-power investments (e.g., Ghana or Nigeria). Central and East Africa, however, may face governance bottlenecks or project misalignment, where RBL funds might be tied to extractive exports rather than domestic grid development. For renewable energy consumption, in Central and East Africa, RBLs are clearly crowding out renewable energy investment, likely due to fossil-backed loan collateralization (oil, gas, minerals). In Southern and West Africa, the weaker negative effects could indicate a more diversified energy mix or partial integration of renewables into broader energy strategies. On the results in Appendix D, RBLs have significant effects in Central and East Africa but an insignificant effect in West Africa.

4 Conclusions and policy implications.

To contribute to the body of knowledge and achieve SDGs 7 and 13, we analyzed the effects of RBLs on energy transition in SSA. We computed Driscoll–Kraay standard errors around the fixed-effects coefficients to account for heteroskedasticity, serial correlation and cross-sectional dependence. To ensure robustness, we estimated a Hausman-Taylor estimator that allows consistent estimation while retaining time-invariant variables. The comparative analysis of our findings offers valuable insights into the dynamics shaping SSA's energy transition. Although the results are integral to achieving SDG 7 and 13, the results reveal complex trade-offs among promoting access to clean cooking fuels, increasing electricity access and promoting renewable energy adoption. They provide an evidence-based understanding of how financial structures, institutional quality, and human capital interact in SSA's evolving energy landscape. The results indicated a pronounced asymmetry in the influence of RBL. RBLs increase access to clean cooking fuels, yet decrease renewable energy consumption and do not affect access to electricity. This pattern suggests that while such loans may support short-term electrification through investments in power infrastructure, they tend to reinforce fossil fuel dependency and carbon-intensive energy pathways. In essence, RBLs may be financing access at the cost of sustainability. It is emphasized that the opaque and extractive nature of RBLs often anchors SSA countries to resource-based energy systems rather than catalyzing renewable transformation.

Governments can integrate RBLs within green finance taxonomies to ensure that future contracts support climate-smart, low-carbon development pathways. Additionally, the encouragement of co-financing models blending RBLs with concessional and private finance can enhance efficiency and accountability.

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Appendix A: measurement, definition and cross-sectional dependence test

	Measurement	CD	CDw+	p1	p99	Skew.	Kurt.	Source
Renewable energy consumption	Renewable energy consumption (% of total final energy consumption)	18.250***	2418.850***	5.6	97	-0.842	2.572	WDI
Access to clean cooking	Access to clean fuels and technologies for cooking (% of population)	48.460***	3730.130***	0	98.1	1.725	5.375	WDI
Access to electricity	Access to electricity (% of population)	137.150***	4217.170***	2.8	99.5	0.539	2.499	WDI
RBL	Countries that have secured resource-backed loans	5.770***	177.280***	0	1	1.902	4.617	Mihalyi et al. (2022)
Resource rent	Total natural resource rents as a percentage of GDP	18.470***	2265.570***	.009	51.844	1.892	7.69	WDI
EnergyGDP	Energy use as a percentage of GDP	44.110***	1278.790***	2.228	44.891	10.541	140.728	WDI
Sustainability	CPIA policy and institutions for environmental sustainability rating (1=low-6=high)	94.730***	2912.230***	1	4.5	-0.705	4.013	QGI(QoG Data)
Stability	Political stability and absence of violence or terrorism	1.050	1859.100***	-2.474	1.064	-0.27	2.343	QGI(QoG Data)
Education	Percentage of gross secondary school enrollment	26.150***	674.560***	6.867	97.075	0.826	3.098	WDI
Block	Subregions in SSA (1=Central Africa,2=East Africa,3=Southern Africa,4=West Africa)							Authors' construct

Appendix B: List of Countries

Angola	Gabon	Niger
Benin	Gambia, The	Nigeria
Botswana	Ghana	Rwanda
Burkina Faso	Guinea	São Tomé and Príncipe
Burundi	Guinea-Bissau	Senegal
Cabo Verde	Kenya	Sierra Leone
Cameroon	Lesotho	Sudan
Central African Republic	Liberia	South Sudan
Chad	Madagascar	Tanzania
Congo, Dem. Rep.	Malawi	Togo
Congo, Rep.	Mali	Uganda

Cote d'Ivoire
Djibouti
Equatorial Guinea
Ethiopia

Mauritania
Mauritius
Mozambique
Namibia

Zambia
Zimbabwe

Appendix C: Sensitivity analyses

	Central Africa								
	Access to clean cooking fuels			Access to electricity			Renewable energy consumption		
	FE	RE	dis kraay	FE	RE	dis kraay	FE	RE	dis kraay
RBL	1.569 (1.014)	2.881 (3.355)	1.569 (0.980)	1.170 (1.520)	-6.682*** (1.908)	1.170 (1.596)	-6.024*** (1.080)	-7.079*** (1.199)	-6.024*** (1.080)
EnergyGDP	0.902** (0.366)	1.893*** (0.536)	0.902** (0.317)	1.310** (0.549)	0.694*** (0.258)	1.310** (0.458)	-0.769* (0.390)	-2.429*** (0.248)	0.769* (0.390)
Education	0.112*** (0.029)	0.339** (0.164)	0.112*** (0.026)	0.233*** (0.044)	0.812*** (0.145)	0.233*** (0.025)	-0.178*** (0.031)	-0.118*** (0.037)	-0.178*** (0.031)
Resource rent	-0.076 (0.048)	0.253 (0.209)	-0.076 (0.048)	-0.033 (0.072)	-0.161 (0.171)	-0.033 (0.039)	0.141*** (0.051)	0.0916* (0.047)	0.141*** (0.051)
FDI	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Stability	-4.632*** (0.824)	-2.061 (5.369)	-4.632** (1.761)	-4.490*** (1.235)	9.448*** (1.453)	-4.490** (1.837)	-0.372 (0.878)	-4.259 (3.241)	-0.372 (0.878)
Sustainability	2.548*** (0.783)	13.35* (7.103)	2.548* (1.380)	3.955*** (1.173)	11.34** (5.076)	3.955** (1.728)	1.366 (0.834)	6.792** (3.329)	1.366 (0.834)
_cons	-13.62*** (4.513)	-66.27** (27.270)	-13.62* (7.066)	-11.66* (6.764)	-21.690 (16.490)	-11.660 (8.886)	88.83*** (4.806)	87.65*** (15.480)	88.83*** (4.806)
N	45	45	45	45	45	45	45	45	45

	East Africa								
	FE	RE	dis kraay	FE	RE	dis kraay	FE	RE	dis kraay
RBL	17.42*** (2.857)	30.75*** (4.797)	17.42*** (2.589)	1.876 (5.483)	11.86 (8.418)	1.876 (3.564)	-6.476*** (1.528)	-10.34*** (1.663)	-6.476*** (1.237)
EnergyGDP	-2.087***	-0.58	-2.087**	2.896**	0.0471	2.896**	-1.096***	-1.987***	-1.096**

	(0.601)	(0.759)	(0.93)	(1.154)	(1.905) 0.682**	(1.294)	(0.321)	(0.230)	(0.473)
Education	0.291***	0.232**	0.291***	0.909***	*	0.909***	-0.249***	-0.199***	-0.249***
	(0.071)	(0.11)	(0.075)	(0.137)	(0.205)	(0.206)	(0.038)	(0.052)	(0.034)
Resource rent	-0.928***	-0.569**	-0.928***	-1.564***	-0.277	-1.564***	0.258**	0.245**	0.258***
	(0.182)	(0.283)	(0.142)	(0.35)	(0.506)	(0.273)	(0.097)	(0.099)	(0.049)
FDI	-2.86e-09**	1.43E-09	-2.86e-09**	-7.59e-09***	2.58E-09	-7.59e-09***	1.82e-09***	3.16E-10	1.82e-09**
	(1.20E-09)	(1.28E-9)	(1.19E-09)	(2.30E-09)	(3.13E-09)	(2.16E-09)	(6.41E-10)	(7.48E-10)	(8.27E-10)
Stability	3.83	-3.039	3.83	3.539	1.962	3.539	0.066	-0.306	0.066
	(2.30)	(1.999)	(2.903)	(4.413)	(3.80)	(4.155)	(1.23)	(1.192)	(1.837)
Sustainability	1.598	-2.217	1.598	-1.053	-1.604	-1.053	0.659	3.585***	0.659
	(1.75)	(3.896)	(1.19)	(3.359)	(4.21)	(1.769)	(0.936)	(1.102)	(1.019)
_cons	19.54***	7.984	19.54**	0.0357	8.741	0.0357	93.04***	89.75***	93.04***
	(6.509)	(16.46)	(7.531)	(12.49)	(23.18)	(11.35)	(3.481)	(4.467)	(3.608)
N	73	73	73	73	73	73	73	73	73

	Southern Africa								
RBL		25.04***			21.89***			-3.473	
		(0.47)			(4.478)			(8.229)	
EnergyGDP	-0.63	0.534***	-0.63	0.884	-0.126*	0.884	-5.935***	-0.830***	-5.935***
	(0.593)	(0.017)	(0.393)	(1.80)	(0.075)	(2.025)	(0.949)	(0.117)	(0.705)
Education	0.016	-0.003	0.016	0.160	0.176*	0.160***	-0.084	-0.166	-0.084**
	(0.031)	(0.009)	(0.013)	(0.095)	(0.101)	(0.050)	(0.050)	(0.190)	(0.036)
Resource rent	0.211	0.045***	0.211*	0.293	0.437**	0.293	-0.071	-0.801***	-0.071
	(0.132)	(0.016)	(0.11)	(0.40)	(0.207)	(0.357)	(0.211)	(0.223)	(0.128)
FDI	-4.05E-11	-6.38E-11	-4.05E-11	7.28e-10**	7.48e-10***	7.28e-10***	-1.66E-10	-2.68E-10	-1.66e-10**
	(9.63E-11)	(1.60E-10)	(7.03E-11)	(2.92E-10)	(1.70E-10)	(1.65E-10)	(1.54E-10)	(3.68E-10)	(6.43E-11)
Stability	0.127	-0.426***	0.127	-1.252	-0.773	-1.252	2.112*	-0.313	2.112**
	(0.617)	(0.069)	(0.409)	(1.869)	(0.522)	(1.214)	(0.986)	(0.708)	(0.771)
Sustainability	2.048*	1.298	2.048	3.035	3.685**	3.035	1.923	-1.367	1.923***
	(0.925)	(2.219)	(1.243)	(2.806)	(1.603)	(2.096)	(1.480)	(2.237)	(0.579)

_cons	8.346 (5.058)	-3.041 (6.606)	8.346** (3.045)	-9.041 (15.34)	-4.294 (3.966)	-9.041 (14.32)	143.3*** (8.091)	106.6*** (7.045)	143.3*** (6.037)
N	17	17	17	17	17	17	17	17	17

	West Africa								
RBL	1.325 (1.793)	1.255 (2.381)	1.325 (1.51)	3.542* (2.091)	3.758** (1.818)	3.542** (1.602)	-0.932 (1.521)	-1.256 (1.668)	-0.932 (1.807)
EnergyGDP	0.952** (0.366)	0.922** (0.432)	0.952** (0.385)	1.116** (0.426)	0.783* (0.452)	1.116** (0.403)	-1.263*** (0.31)	-0.920* (0.511)	-1.263*** (0.405)
Education	0.201*** (0.058)	0.211 (0.191)	0.201* (0.107)	0.514*** (0.068)	0.525*** (0.123)	0.514*** (0.066)	-0.261*** (0.049)	-0.280*** (0.104)	-0.261*** (0.078)
Resource rent	0.077 (0.164)	0.085 (0.119)	0.077 (0.156)	-0.126 (0.191)	-0.24 (0.274)	-0.126 (0.147)	-0.397*** (0.139)	-0.318** (0.141)	-0.397** (0.184)
FDI	-1.46e-09*** (4.12E-10)	-1.48e-09*** (2.29E-10)	-1.46e-09** (5.35E-10)	5.15E-10 (4.80E-10)	7.22E-10 (6.16E-10)	5.15E-10 (3.31E-10)	4.43E-10 (3.49E-10)	3.67E-10 (4.86E-10)	4.43E-10 (3.19E-10)
Stability	-0.754 (1.333)	-0.477 (2.236)	-0.754 (0.89)	2.567 (1.554)	2.849 (1.889)	2.567* (1.336)	2.185* (1.13)	1.254 (1.429)	2.185* (1.068)
Sustainability	3.958** (1.51)	4.045 (2.541)	3.958*** (1.027)	4.304** (1.761)	4.543* (2.373)	4.304*** (1.018)	-1.842 (1.281)	-2.314 (3.245)	-1.842 (1.609)
_cons	-19.21*** (6.259)	-19.56*** (6.949)	-19.21*** (6.03)	-6.295 (7.296)	-5.281 (11.63)	-6.295 (7.737)	103.9*** (5.309)	101.5*** (9.486)	103.9*** (9.197)
N	99	99	99	99	99	99	99	99	99

Note: Standard errors are in brackets. *p<0.10, **p<0.05, ***p<0.01.

Appendix D: Robustness of the sensitivity analyses

Note: Standard errors are in brackets.*p<0.10,**p<0.05,***p<0.01.

	Central Africa			East Africa			West Africa		
	Clean cooking fuels	Access to electricity	Renewable energy consumption	Clean cooking fuels	Access to electricity	Renewable energy consumption	Clean cooking fuels	Access to electricity	Renewable energy consumption
RBL	3.747*** (1.281)	3.830*** (1.314)	-6.616*** (1.024)	22.281*** (2.177)	3.707 (3.720)	-9.001*** (1.233)	-1.855 (1.830)	0.435 (2.383)	0.133 (1.901)
EnergyGDP	-0.475*** (0.179)	-0.210 (0.184)	-1.021*** (0.139)	-0.726** (0.353)	0.040 (0.442)	-1.004*** (0.201)	1.151*** (0.330)	1.034** (0.407)	-0.869** (0.357)
FDI	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)	-0.000** (0.000)	0.000* (0.000)	-0.000* (0.000)	0.000* (0.000)	-0.000** (0.000)
Political stability	-3.704** (1.512)	-2.968* (1.552)	-1.705 (1.165)	-0.254 (1.511)	-0.081 (2.526)	3.287*** (0.855)	-3.071*** (1.036)	1.432 (1.346)	3.164*** (1.078)
Education	0.251*** (0.041)	0.370*** (0.042)	-0.169*** (0.033)	0.316*** (0.063)	0.958*** (0.102)	-0.234*** (0.036)	0.155*** (0.041)	0.667*** (0.053)	-0.239*** (0.043)
Resources rents	0.005 (0.067)	0.082 (0.069)	0.026 (0.054)	-1.114*** (0.195)	-1.411*** (0.325)	0.320*** (0.111)	0.218 (0.175)	-0.419* (0.226)	-0.231 (0.183)
Country	0.952 (2.384)	1.494 (2.565)	1.429** (0.629)	-0.418 (1.949)	0.094 (0.680)	-0.415 (1.227)	0.080 (0.403)	0.152 (0.426)	0.037 (0.512)
_cons	9.323 (25.628)	5.948 (27.560)	84.194*** (7.085)	31.689 (58.956)	7.789 (23.327)	106.475*** (37.050)	-11.395 (11.353)	-4.466 (12.476)	90.459*** (13.998)
sigma_u	15.317	30.875	20.102	29.057	30.875	20.102	18.583	14.400	15.117
sigma_e	6.080	4.374	4.079	1.706	4.374	4.079	3.170	3.017	3.916
rho	0.864	0.980	0.960	0.997	0.980	0.960	0.972	0.958	0.937
N	55	55	55	97	97	97	97	97	97